

Problem 1:

You are operating a batch reactor and the reaction $A \rightarrow B$ is first order, liquid phase, and exothermic. An inert coolant is added to the reaction mixture to control the temperature. The temperature is kept constant by varying the flow rate of the coolant.

- (a) Calculate the flow rate of the coolant 2h after the start of the reaction.
- (b) It is proposed that rather than feeding a coolant to the reactor, a solvent be added that can be easily boiled off, even at moderate temperatures. The solvent has a heat of vaporization of 1000Btu/lb and initially there are 25 lb mol of A placed in the tank. The initial volume of solvent and reactant is 300 ft³. Determine the solvent evaporation rate as a function of time. What is the rate at the end of 2h?

Information:

Temperature of reaction: 100 F

Value of k at 100 F: 1.2×10^{-4}

Temperature of coolant: 80F

Heat capacity of all components: 0.5 Btu/lb.F

Density of all component: 50 lb/ft³

$\Delta H_{Rx}^0 = -25,000$ Btu/lb mol

Initially: vessel contains A only. $C_{A0} = 0.5$ lb mol/ft³. Initial volume = 50 ft³.